SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	B.Tech AI/DS	**Date:**	

Code of Conduct

I Nirmal Kasi 192424248 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Nirmal

Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications.	Q1
Syntax-Driven Semantic Analysis	Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.	Q2
First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis	Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.	Q3

Annexure A – Industry Problem Solving Task

1. AI-Powered Insurance Claim Processing System

An insurance company is developing an NLP-based system to automatically process customer insurance claims. The system uses **Context-Free Grammar (CFG)** to parse customer statements and extract information such as the claimant, damaged object, cause of damage, and claim amount.

However, the system faces the following problems:

- Ambiguous sentence structures
- Difficulty distinguishing noun and prepositional phrase attachment
- Subject–verb agreement errors
- Inefficient parsing of lengthy customer descriptions
- Difficulty handling conversational and incomplete input

Example customer statement:

"I reported the damage to the car after the accident."

The sentence can produce different interpretations depending on whether **"to the car"** is associated with *reported* or *damage*.

Tasks:

a) Analyze the structural ambiguity in the given sentence using CFG and illustrate the possible interpretations.

b) Evaluate the limitations of a basic CFG in handling ambiguity, agreement, and long customer-generated insurance statements.

c) Design an improved parsing approach using **PCFG, Feature Structures, and Earley Parsing** to resolve ambiguity and efficiently process the claim.

d) Justify how the proposed architecture can improve the accuracy, scalability, and reliability of automated insurance claim processing.

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2. Intelligent Railway Ticket Booking Assistant

A railway company is developing a conversational AI assistant for ticket booking. The system currently uses **top-down parsing** to interpret passenger requests.

The system encounters:

- Excessive backtracking
- Ambiguous attachment of phrases
- Incomplete passenger requests
- Long-distance dependencies
- Slow response for complex queries

Example passenger request:

"Book two tickets from Chennai to Delhi for my parents with AC sleeper."

The system sometimes incorrectly associates **"with AC sleeper"** with the passengers instead of the ticket/class information.

Tasks:

a) Analyze the possible syntactic interpretations of the given passenger request using CFG.

b) Evaluate why top-down parsing may result in excessive backtracking and inefficient processing for conversational ticket-booking queries.

c) Analyze how **Earley Parsing** can handle ambiguity, incomplete input, and different possible parse structures more effectively.

d) Compare **Top-Down Parsing, CFG-based parsing, and Earley Parsing** in terms of ambiguity handling, computational efficiency, partial-input processing, and suitability for real-time railway booking systems.

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3. AI-Based Legal Contract Analysis System

A legal technology company is developing an NLP system to analyze contracts and automatically identify:

- Parties involved
- Obligations

- Conditions
- Deadlines
- Actions
- Exceptions

The existing system uses a basic CFG parser. However, it produces incorrect interpretations when processing long legal sentences containing relative clauses, passive constructions, and multiple prepositional phrases.

Example contract sentence:

"The supplier who delivered the equipment to the company must replace the defective components within thirty days."

The system sometimes incorrectly identifies the **company** as the entity responsible for delivering the equipment.

Tasks:

a) Analyze the syntactic structure of the given sentence using CFG and identify possible sources of ambiguity.

b) Evaluate why basic CFG is insufficient for accurately interpreting complex legal sentences containing relative clauses and nested structures.

c) Design an NLP architecture integrating **CFG, PCFG, Feature Structures, and Earley Parsing** to improve syntactic and semantic interpretation.

d) Develop a step-by-step processing workflow showing how the system can extract:

- Supplier
- Action
- Equipment
- Company
- Obligation
- Deadline

and justify how the proposed approach improves automated contract analysis.

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4. AI-Based E-Commerce Product Search System

An e-commerce company is developing a conversational product-search system. Customers enter natural-language queries rather than selecting predefined filters.

The current system uses CFG but faces difficulties with:

- Ambiguous phrase attachment
- Coordination
- Missing words
- Informal customer queries
- Long product descriptions
- Real-time search requirements

Example user query:

"Find laptops with a touchscreen for students under ₹60,000."

The system may incorrectly associate **"for students"** with the touchscreen rather than the laptop.

Tasks:

a) Analyze the syntactic ambiguity in the given query and construct possible parse interpretations using CFG.

b) Evaluate the limitations of CFG in handling informal, elliptical, and ambiguous e-commerce search queries.

c) Propose an improved parsing architecture using **PCFG, Feature Structures, and Earley Parsing** to identify product attributes, user requirements, and price constraints.

d) Justify how the proposed solution can improve search precision, response time, and customer experience in a large-scale e-commerce platform.

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5. Intelligent Airline Voice Assistant

An airline is developing a real-time voice assistant that allows passengers to modify flight bookings using spoken commands.
The system currently uses a conventional top-down parser. It experiences:

- Backtracking
- Ambiguous attachment
- Speech recognition errors
- Incomplete commands
- Delayed responses
- Difficulty interpreting corrections made by users

Example spoken command:
"Change my flight to Mumbai tomorrow with two passengers."
Depending on the parse, **"with two passengers"** may be incorrectly interpreted as an attribute of the destination rather than the number of passengers.
The passenger may also say:
"Change my flight to Mumbai tomorrow... actually, make that Bangalore."
Tasks:
a) Analyze the possible syntactic structures and ambiguities in the given voice command.
b) Evaluate the limitations of top-down parsing when processing incomplete, corrected, and ambiguous spoken commands in real time.
c) Design a parsing architecture using **Earley Parsing, PCFG, and Feature Structures** to support partial input, ambiguity resolution, and agreement constraints.
d) Develop a processing workflow showing how speech input can be converted into a structured booking request and justify the proposed method for real-time airline applications.

Q1. AI-Powered Insurance Claim Processing System

a)

Sentence: **“I reported the damage to the car after the accident.”**

Possible interpretations:

1. **“to the car” modifies “damage”**
   Meaning: I reported **damage to the car** after the accident.
2. **“to the car” modifies “reported”**
   Meaning: I reported **something to the car**, which is grammatically less natural.

CFG can generate multiple parse trees because the prepositional phrase **“to the car”** can attach to different parts of the sentence.

b)

Basic CFG has the following limitations:

- Cannot effectively resolve structural ambiguity.
- Does not naturally enforce subject–verb agreement.
- Produces multiple possible parses without ranking them.
- Becomes inefficient for long sentences.
- Handles conversational and incomplete inputs poorly.
- Has difficulty with complex noun and prepositional phrase attachments.

c)

Input → CFG Parsing → Feature Structures → PCFG Ranking → Earley Parsing → Best Parse → Claim Information Extraction

- **CFG:** Generates possible syntactic structures.
- **Feature Structures:** Enforce agreement such as number and grammatical features.
- **PCFG:** Assigns probabilities to competing parses and selects the most likely interpretation.
- **Earley Parsing:** Efficiently handles ambiguity, incomplete input, and complex sentence structures.

d)

The architecture improves:

- **Accuracy:** PCFG ranks interpretations and Feature Structures enforce grammatical constraints.
- **Scalability:** Earley Parsing efficiently handles long and complex sentences.
- **Reliability:** Semantic roles such as claimant, damaged object, cause, and claim amount can be extracted from the most appropriate parse.

Q2. Intelligent Railway Ticket Booking Assistant

a)

Sentence:

“Book two tickets from Chennai to Delhi for my parents with AC sleeper.”

Possible structures:

1. **“with AC sleeper” → ticket/class**
   - 2 tickets
   - From Chennai
   - To Delhi
   - For parents
   - Class = AC Sleeper
2. **“with AC sleeper” → parents**
   - Parents are incorrectly associated with “AC sleeper”.

The intended interpretation is that **AC sleeper is the ticket/class information**.

b)

Top-down parsing starts from the grammar's start symbol and predicts possible structures. When an incorrect prediction is made, it may need to **backtrack** and try another possibility.

For complex conversational queries, this causes:

- Excessive backtracking
- Increased computation
- Slow response
- Difficulty handling ambiguous attachments
- Poor handling of incomplete requests
- Difficulty with long-distance dependencies

c)

Earley Parsing is better because it can:

- Handle ambiguous grammar.
- Process incomplete input.
- Maintain multiple possible parse states.
- Avoid unnecessary repeated parsing through chart-based processing.
- Handle complex and long-distance structures efficiently.

d)

Feature	Top-Down Parsing	CFG Parsing	Earley Parsing
Ambiguity	Poor	Generates alternatives	Handles alternatives effectively
Efficiency	Can suffer from backtracking	Depends on implementation	Efficient for general CFG
Incomplete input	Poor	Limited	Good
Complex queries	Less suitable	Moderate	Highly suitable
Real-time assistant	Less suitable	Moderate	Better

Conclusion: Earley Parsing is the most suitable for a real-time railway booking assistant because it handles ambiguity, incomplete input, and complex structures more effectively.

Q3. AI-Based Legal Contract Analysis System

a)

Sentence:

“The supplier who delivered the equipment to the company must replace the defective components within thirty days.”

Structure:

- Main subject: **The supplier**
- Relative clause: **who delivered the equipment to the company**
- Main action: **must replace**
- Object: **defective components**
- Deadline: **within thirty days**

Possible ambiguity occurs because **“to the company”** is a prepositional phrase that must be correctly attached within the relative clause.

Correct relationship:

Supplier → delivered → Equipment → to → Company

The supplier is responsible for delivering the equipment, not the company.

b)

Basic CFG is insufficient because:

- It does not adequately resolve ambiguity.
- It struggles with nested structures.
- Relative clauses create complex dependencies.
- Passive constructions can confuse semantic roles.
- Multiple prepositional phrases can have different attachments.
- Long legal sentences can produce many possible parse trees.

c)

Legal Contract → Tokenization → CFG → Feature Structures → PCFG → Earley Parsing → Semantic Role Analysis → Entity/Obligation Extraction

- **CFG:** Generates grammatical structures.
- **Feature Structures:** Enforce grammatical constraints.
- **PCFG:** Ranks competing interpretations.
- **Earley Parsing:** Efficiently handles complex and incomplete structures.
- **Semantic analysis:** Identifies entities, actions, obligations, and deadlines.

d)

Information	Extracted Value
Supplier	The supplier
Action	Delivered
Equipment	The equipment
Company	The company
Obligation	Replace defective components
Deadline	Within thirty days

Processing:

```
Sentence
↓
Syntactic Parsing
↓
Relative Clause Identification
↓
PP Attachment Resolution
↓
Semantic Role Assignment
↓
Entity & Obligation Extraction
↓
Deadline Extraction
```

The proposed architecture improves contract analysis by correctly identifying **who performed an action, what was affected, who received it, what obligation exists, and when it must be completed**.

Q4. AI-Based E-Commerce Product Search System

a)

Query:

“Find laptops with a touchscreen for students under ₹60,000.”

Possible interpretations:

1. **Correct:**
   `Laptops → with touchscreen → for students → under ₹60,000`
2. **Incorrect:**
   `Touchscreen → for students`

The intended meaning is that **the laptops are for students**, not that the touchscreen is for students.

b)

CFG limitations:

- Difficulty resolving phrase attachment.
- Poor handling of informal queries.
- Difficulty with missing or elliptical words.
- Difficulty handling coordination.
- Inefficient for long descriptions.
- Does not naturally rank competing interpretations.
- Limited suitability for real-time search.

c)

User Query → CFG → Feature Structure Analysis → PCFG Ranking → Earley Parsing → Semantic Extraction → Search Filters

Extracted information:

- **Product:** Laptop
- **Feature:** Touchscreen

- **Target user:** Students
- **Price constraint:** Below ₹60,000

Feature Structure:

```
Product = Laptop
Feature = Touchscreen
User = Student
Price < ₹60,000
```

d)

The proposed architecture improves:

- **Search precision:** Correctly separates product attributes and user requirements.
- **Response time:** Earley Parsing efficiently processes complex queries.
- **Accuracy:** PCFG selects the most probable interpretation.
- **Constraint handling:** Feature Structures maintain relationships between product, features, users, and price.
- **Customer experience:** Users receive results matching their actual query instead of whatever grammatical interpretation the parser hallucinated into existence.

Q5. Intelligent Airline Voice Assistant

a)

Command:

"Change my flight to Mumbai tomorrow with two passengers."

Possible interpretations:

1. **Correct:**
   Destination = Mumbai
   Date = Tomorrow
   Passengers = 2
2. **Incorrect:**
   "with two passengers" is associated with Mumbai as a destination attribute.

Correction:

"Change my flight to Mumbai tomorrow... actually, make that Bangalore."

The final destination should be **Bangalore**, replacing Mumbai.

b)

Top-down parsing has difficulty with:

- Excessive backtracking
- Ambiguous phrase attachment
- Speech recognition errors
- Incomplete commands
- User corrections

- Delayed responses
- Real-time processing

For example, the system must understand that **“actually, make that Bangalore”** corrects the previously stated destination rather than adding another destination.

c)

Speech Input → Speech Recognition → Earley Parsing → Feature Structures → PCFG Ranking → Correction Detection → Semantic Interpretation → Dialogue State Update → Booking Request

- **Earley:** Handles partial and ambiguous input.
- **PCFG:** Ranks possible interpretations.
- **Feature Structures:** Enforce constraints such as number and grammatical agreement.
- **Correction Detection:** Replaces Mumbai with Bangalore when the user corrects the destination.

d)

1. Receive spoken command.
2. Convert speech to text.
3. Identify entities such as Mumbai, tomorrow, and two passengers.
4. Generate possible parse structures using Earley Parsing.
5. Apply Feature Structures to enforce constraints.
6. Use PCFG to select the most probable interpretation.
7. Detect user corrections.
8. Update the destination from **Mumbai → Bangalore**.
9. Construct the structured booking request.
10. Send the validated request to the booking system.

Final structured request:

```
Action = Change Flight
Destination = Bangalore
Date = Tomorrow
Passengers = 2
```

This approach provides better ambiguity handling, supports incomplete and corrected speech, and reduces response delays in real-time airline applications.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Q. No. / Criteria Level	Understanding of Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analysis	Professionalism	Sub. Total	Total %
1	4	3	3	3	3	3	19	94
2	3	3	3	3	3	3	18	
3	3	3	3	4	3	3	19	
4	3	4	3	3	3	3	19	
5	3	3	4	3	3	3	19	

Faculty In-charge	Course Coordinator	Program Director
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Date: ____________________	Date: ____________________	Date: ____________________